TSA 2022

Problem Solving

Questions taken from the TSA Oxford 2022 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2022 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

Questions © Cambridge University Press & Assessment 2022. Reproduced from the TSA Oxford 2022 Section 1 question paper for practice purposes.

1 (Q2) A quiz consists of 7 rounds, each of which contains 20 questions. Each question is worth one point. At the end of the quiz, but before any of the answers or scores are revealed, each team is allowed to select one of the rounds and the score for that round is tripled. In addition, the scores for the rounds immediately before and after the chosen round are doubled. The winning team at last week's quiz had the following scores in each of the rounds.

round	1	2	3	4	5	6	7
score	14	16	6	20	10	18	4

The winning team's final score was 136.

Which round did the winning team choose to have the score tripled?

A round 2

B round 3

C round 4

D round 5

E round 6

2 (Q6) Rob has two different mixtures of apple juice and orange juice.

mixture 1 40% apple juice and 60% orange juice

mixture 2 70% apple juice and 30% orange juice

Rob wants to use these two mixtures to make 6 litres of a juice mixture which is 50% apple juice and 50% orange juice.

How much of mixture 2 does Rob need to use?

A 1 litre

B 2 litres

C 3 litres

D 4 litres

E 5 litres

© Cambridge University Press & Assessment 2022

3 (Q7) The extension numbers needed to contact eight different departments in a university depend on whether the caller is calling as a student, parent, lecturer, journalist or researcher. The appropriate numbers are shown below.

	student	*parent*	*lecturer*	*journalist*	*researcher*
Accounts	387	387	661	387	661
Administration	117	387	117	232	232
Enquiries	239	387	387	661	387
Complaints	558	239	117	387	117
Quality Assurance	239	239	117	558	239
Marketing	387	239	232	661	232
Public Relations	239	117	387	239	232
Human Resources	239	239	661	117	239

Fran is a lecturer and regularly needs to contact all of the departments in the university.

How many of the extension numbers are **never** needed by Fran?

A 0

B 1

C 2

D 3

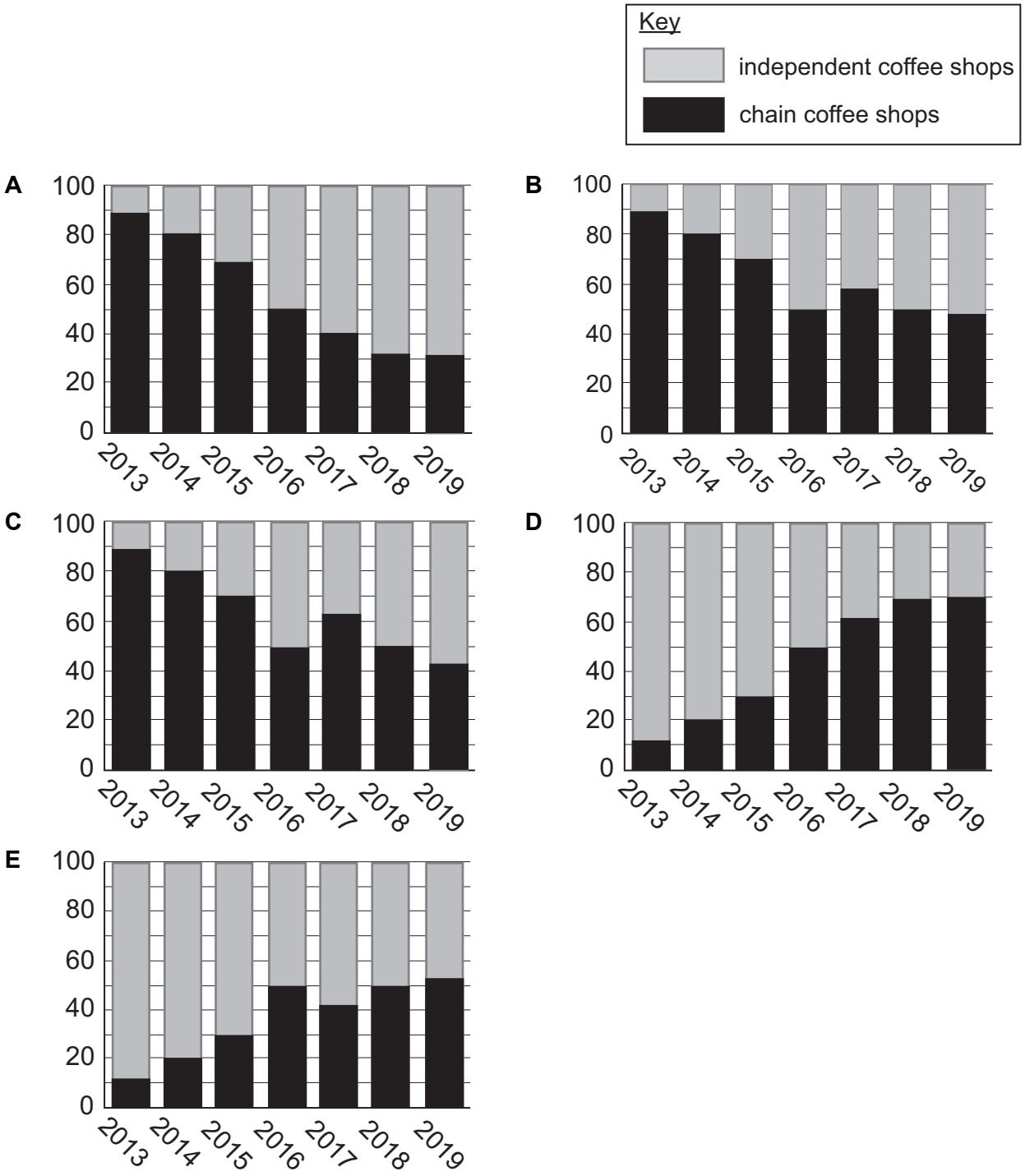
E 4

© Cambridge University Press & Assessment 2022

4 (Q8) Over the past decade the popularity of coffee has soared in the United Kingdom. As a result, the number of coffee shops across the UK has risen. The table below shows a breakdown of the number of chain and independent coffee shops in East London between 2013 and 2019.

	number of coffee shops in East London						
	2013	2014	2015	2016	2017	2018	2019
chain	16	20	35	45	70	100	130
independent	2	5	15	45	50	100	140
total	18	25	50	90	120	200	270

Which graph shows the information in the table above?



© Cambridge University Press & Assessment 2022

5 (Q12) The shampoo that I normally buy costs £1.96 for a 500 ml bottle and always lasts for 40 days.

Last month I decided to try a new brand that is 36p cheaper and contains 550 ml. However, because of the amount that I have had to use to wash my hair properly, it has run out after only 25 days.

Based upon my experience, by how many pence per day is my normal shampoo better value than the new brand?

A 1.5

B 2.4

C 4.9

D 6.4

E 9.5

6 (Q13) Kate and Tek are choosing a cottage to rent for their family holiday. They have searched online and reduced their choice to five cottages. The facilities offered by these cottages are shown in the following table.

cottage	*number of bedrooms*	*large garden*	*wifi*	*parking*	*pets allowed*	*distance to nearest store*	*cost per week*
Acorns	3	no	yes	no	yes	2 km	$460
Beeches	3	yes	yes	no	yes	1 km	$490
Chestnuts	4	no	yes	yes	no	2 km	$510
Denders	4	yes	yes	yes	no	4 km	$500
Eglers	2	no	yes	yes	no	2 km	$450

Ideally, Kate and Tek would like a cottage with at least three bedrooms, a large garden, wifi, parking, and they would like to be within 3 km of a store. They realise that it is unlikely that all their wishes will be met, so they decide to compromise. They insist that the cottage must have at least three bedrooms, but they will be happy if at least three of their other wishes are satisfied. If more than one cottage is suitable, then they will choose the one that is the cheapest to rent.

Which cottage will Kate and Tek choose?

A Acorns

B Beeches

C Chestnuts

D Denders

E Eglers

© Cambridge University Press & Assessment 2022

7 (Q14) Alex and Anita play a game where they need to form words from letters. The value of each letter is a whole number of points (which can be 0). Alex and Anita earn points by adding together the value of the letters in the words they form.

The value of the last three words they formed is shown in the table below.

word	*value*
MOAN	7
MOON	8
NOON	10

Anita has just formed the word ANNA.

What is the value of the word ANNA?

A 4

B 6

C 8

D 10

E 12

8 (Q18) Sean is taking his car to Sunny Island. He lives 38 miles away from the mainland terminal where the seaport and airport are based. He has researched the travel costs and come up with two possible options. The ferry ticket (car and passenger) is $80. Alternatively, a plane ticket is $50 and the seaport company will transport his car (port to port) for $100. Fuel costs are $1.50 per mile.

How much money will Sean save by taking the cheaper option?

A $13

B $20

C $30

D $70

E $127

© Cambridge University Press & Assessment 2022

9 (Q19) A youth club is open six evenings every week. There are nine volunteers who run the club. Each evening three of the volunteers are on duty.

When the rota is drawn up for any week, the following conditions must be met:

- Each volunteer has two evenings of duty, but never two consecutive evenings.
- No two volunteers are on duty together twice in the same week.

Next week's rota has just been published, as follows:

Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*	*Saturday*
Mel	Eve	Jan	Mel	Eve	Fay
Rod	Fay	Leo	Pat	Jan	Leo
Sam	Tim	Sam	Tim	Rod	Pat

Fay wants to swap her Tuesday duty with one of the others to be on duty on Monday or Thursday instead.

Who is the only volunteer on the rota for Monday or Thursday that Fay could swap with if the conditions described above are still to be met?

A Mel

B Pat

C Rod

D Sam

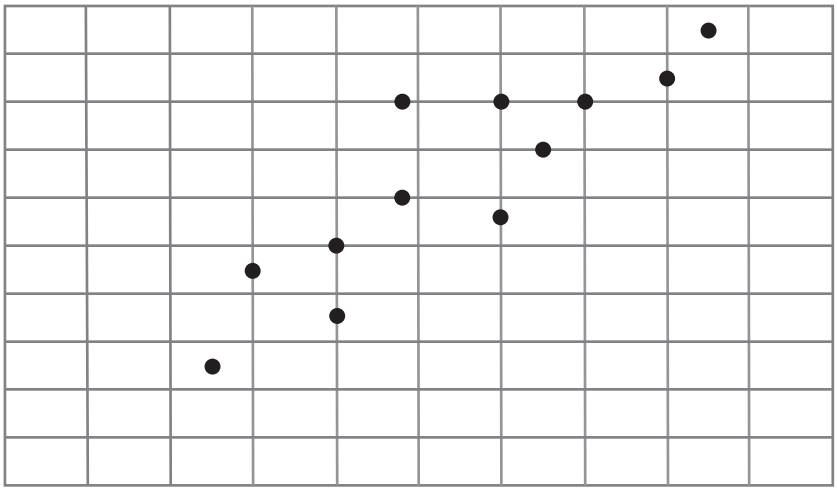
E Tim

© Cambridge University Press & Assessment 2022

10 (Q20) Students studying geography at college were tested at the start of the year and at the end of the year. Their marks are shown in the following table.

student	Ffion	Gary	Huw	Ivy	Jim	Ken	Lee	Mei	Naz	Oli	Pete	Quin
starting mark	60	48	40	30	70	48	65	60	25	85	40	80
end mark	56	50	35	45	80	60	70	80	25	95	50	85

The following scatter graph shows the marks for the 12 students, but one of the points has been incorrectly plotted.



Which student has a mark that is incorrectly plotted on the scatter graph?

A Ffion

B Gary

C Huw

D Ken

E Mei

11 (Q24) Iman and Maya ran a 100 metre race. Iman started 2 seconds after Maya and finished 3 seconds before her.

If each girl ran at a constant speed throughout the race, and Iman's speed was 25% faster than Maya's, for how many seconds did Iman run?

A 15 seconds

B 18 seconds

C 20 seconds

D 22 seconds

E 25 seconds

© Cambridge University Press & Assessment 2022

12 (Q25) Five popular makes of fishcake are shown in the table below, with comparisons of their ingredients, the weight of each type and the cost for a pack of two.

type	*ingredients in fishcake*			*weight of each fishcake*	*cost of pack of 2 fishcakes*
	fish	*potato*	*coating*		
Arctic	45%	40%	15%	100 g	£3.00
Banquet	40%	50%	10%	100 g	£2.50
Chilco	55%	35%	10%	150 g	£5.00
Dyner	50%	35%	15%	150 g	£4.00
Evertop	35%	45%	20%	50 g	£1.00

Ester wants to buy the fishcakes that contain the greatest amount of fish per pound spent.

Which make of fishcake should Ester buy?

A Arctic

B Banquet

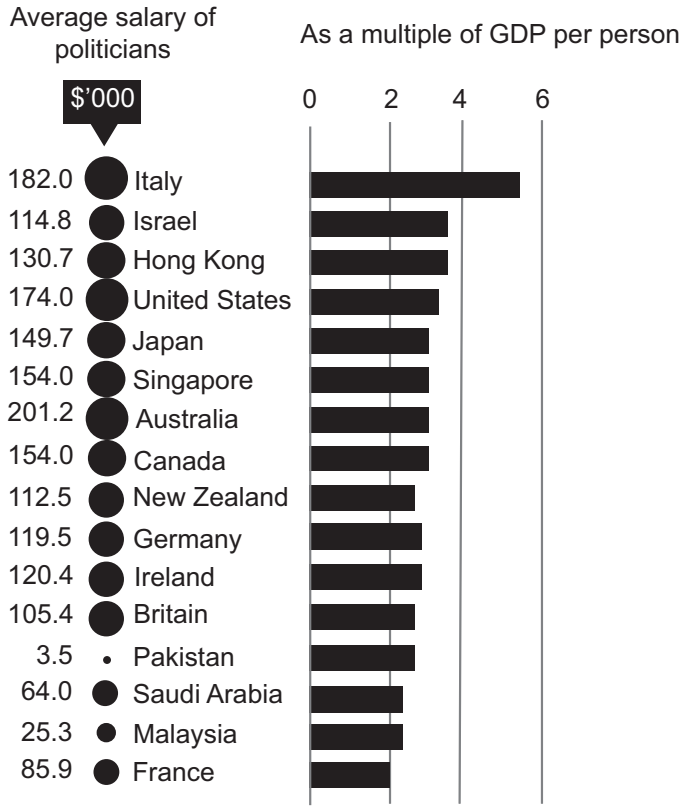
C Chilco

D Dyner

E Evertop

© Cambridge University Press & Assessment 2022

13 (Q26) The following chart shows the average salary of politicians in selected countries in absolute terms on the left (in thousands of dollars) and as a multiple of their country's gross domestic product (GDP) per person on the right.



What is the difference between the highest and lowest average salaries of politicians in the countries listed above?

A $96 100

B $115 300

C $175 900

D $178 500

E $197 700

© Cambridge University Press & Assessment 2022

14 (Q30) In a prize draw, tickets are printed with 3-digit numbers from 001 to 999. Contestants pay to take a ticket at random. A major prize is awarded for all tickets where the product of the 3 digits on the ticket drawn is 8 and a minor prize is awarded when the product is 12. For example 143 wins a minor prize because $1 \times 4 \times 3 = 12$.

What is the maximum number of major prizes which could be won?

A 7

B 9

C 10

D 13

E 14

15 (Q31) The table shows the cost of hot drinks in a local café.

hot drinks		
tea	*coffee*	*chocolate*
£1.50	£2.50	£3.00

Belle bought a number of hot drinks, including at least one of each tea, coffee and chocolate. She was charged £32.00.

If she bought the same number of teas as coffees, what was the maximum number of hot drinks that Belle could have bought for the £32.00?

A 9

B 10

C 12

D 13

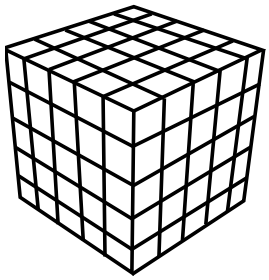
E 14

© Cambridge University Press & Assessment 2022

16 (Q32) I have a wooden cube with edges that are 5 cm long. I paint it in the following way:

- one pair of opposite faces is red
- one pair of opposite faces is green
- one pair of opposite faces is blue

Now I cut the cube into small cubes, each of whose edges are 1 cm long, in the following manner:



How many of the small cubes have one green face, one blue face, but no red face?

A 6

B 8

C 12

D 16

E 24

© Cambridge University Press & Assessment 2022

17 (Q36) Grandma has spent £60 on five books which she is intending to give to her three grandchildren, Miya, Harriet and Mehran. She is going to give the books in a way that relates the cost to the time they normally spend reading (i.e. spend the most on the child who reads the most). Miya reads for an hour a day. Harriet reads for an hour a day on Monday to Thursday and 2 hours a day on Friday to Sunday. Mehran doesn't read on weekdays but does an hour and a half on each of Saturday and Sunday. The details of the books are shown in the table below.

title	*cost*
Hungry Dinosaurs	£8
The Celebrated Six	£9
Fly Away	£13
Come Back	£14
The Biraffe	£16

Which one of the following statements is true?

A Harriet was given 'The Celebrated Six'.

B Harriet was given 'Fly Away'.

C Mehran was given 'The Biraffe'.

D Miya was given 'Come Back'.

E Miya was given 'Hungry Dinosaurs'.

18 (Q37) In a ball game points are accumulated by different combinations of four winning shots called a chop, a creamer, a glink and a yip. Playing the game for the first time, the only knowledge I have about the scoring system is that a chop is worth one point more than a creamer, and a creamer is worth twice as many points as a glink. I lose narrowly by 47 points to 50, scoring 3 chops, 4 yips, 3 creamers and 4 glinks. My victorious opponent scored 5 chops, 3 yips, 3 creamers and 2 glinks.

Which one of the following statements about the scoring system is the only one that is true?

A A creamer is worth more points than a chop.

B Two chops are worth less than three yips.

C A glink is worth 2 points.

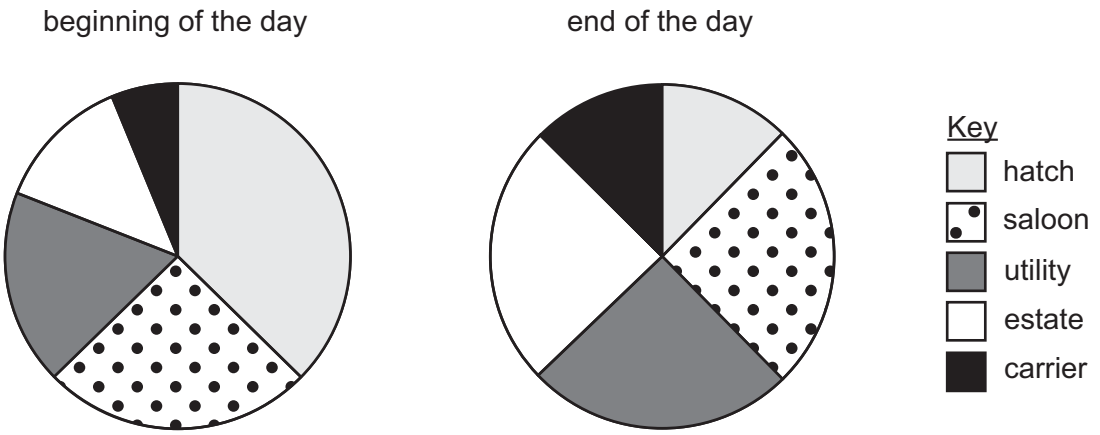
D Three yips are worth more than three creamers.

E A chop is worth 6 points.

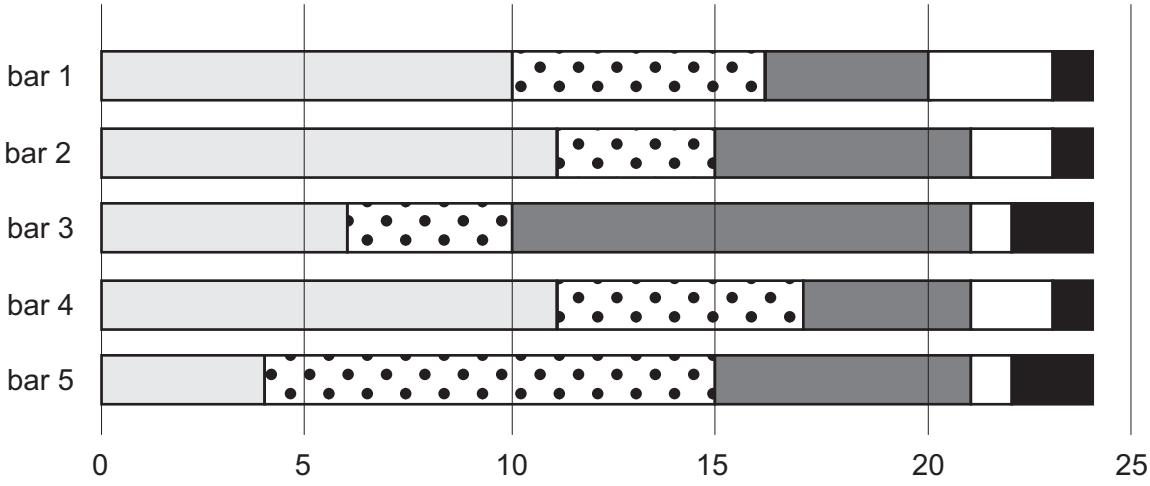
© Cambridge University Press & Assessment 2022

19 (Q38) At the beginning of the day, a showroom had 32 vehicles for sale. The vehicles were advertised in five categories: hatchback, saloon, utility, estate car and people carrier. By the end of the day, 24 vehicles had been sold.

The first chart shows a comparison of the numbers of vehicles in each category in the showroom at the beginning of the day, and the second chart shows a comparison of the numbers in the same categories in the showroom at the end of the day.



Which one of the following shows a comparison of the numbers of cars in each category that were sold?



A bar 1

B bar 2

C bar 3

D bar 4

E bar 5

© Cambridge University Press & Assessment 2022

20 (Q42) In my currency, there are six types of coin in circulation:

200p 100p 50p 20p 10p 5p

I start with two coins. I visit a shop and purchase two bars of chocolate and pay using the two coins. My change consists of two coins.

The next day I re-visit the shop and purchase another one bar of chocolate with that change. My new change, again, consists of two coins.

The six coins I possess during these two days are all different. I only have two coins at any particular time. The price of a bar of chocolate does not change.

What is the price of a bar of chocolate?

A 15p

B 25p

C 30p

D 55p

E 60p

© Cambridge University Press & Assessment 2022

21 (Q43) In a game, players take it in turns to place tokens onto squares of a grid. The squares on the grid are coloured white and grey. When a player places a token they score one point for every token on an adjacent square (either horizontally or vertically). For every adjacent square that is white an extra point is scored.

The diagram below shows a position in one game. The number on each square shows the number of tokens currently placed on that square:

3	1	5	1	2
1	2	0	4	5
2	1	2	3	1
1	2	2	5	2
5	4	3	2	4

What is the maximum number of points that could be scored when the next token is placed on the board?

A 10

B 11

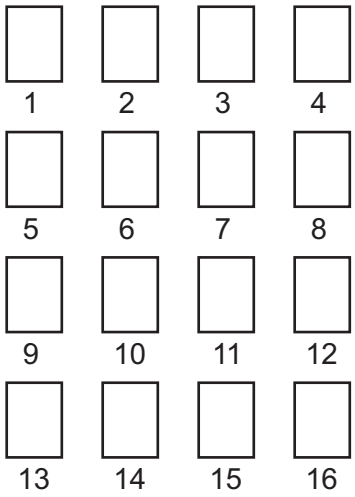
C 12

D 13

E 14

© Cambridge University Press & Assessment 2022

22 (Q44) In *Cat and Mouse*, a game for two players, 16 cards are shuffled and then placed face down on a table as shown below:



[The numbers below the cards are for identification purposes.]

Cats are pictured on 8 of the cards and mice on the other 8. Both players pick up two cards at each turn, taking care not to show the cards to the other player. If the two cards are a cat and a mouse, the player **keeps the cards** and scores a point, but does not reveal which was the cat and which was the mouse. If, however, the two cards are both cats or both mice, the player must put the cards back in the same places.

My sister and I are currently playing a game, and I am leading by 3 points to 2. The game has progressed so far as follows:

me	cards 2 & 13	cat and mouse pair	1 point
sister	cards 7 & 16	cards replaced	
me	cards 5 & 11	cat and mouse pair	1 point
sister	cards 10 & 12	cards replaced	
me	cards 3 & 8	two mice; cards replaced	
sister	cards 7 & 14	cat and mouse pair	1 point
me	cards 1 & 15	cat and mouse pair	1 point
sister	cards 8 & 10	cat and mouse pair	1 point

It is my turn again and I know for certain that the two cards I am going to pick up will be a cat and mouse pair.

Which two cards am I going to pick up?

A 3 & 4

B 3 & 6

C 3 & 9

D 3 & 12

E 3 & 16

© Cambridge University Press & Assessment 2022

23 (Q46) On Sports Day at St Jude's College points are awarded in each event as follows:

1st place	10 points
2nd place	6 points
3rd place	3 points
4th place	1 point

Each student may enter a maximum of three events.

What is the lowest score that is **not** possible for a student to achieve?

A 15

B 22

C 24

D 27

E 31

24 (Q48) My three friends bought food at the supermarket. When I asked them, they did not remember the prices of the items, only the final amount they paid.

- Hector paid $3.50 for 3 bread rolls and 1 apple.
- Lee paid $8.00 for 1 apple and 4 chocolate bars.
- Olga paid $3.50 for 1 bread roll and 2 chocolate bars.

I want to buy 1 bread roll, 1 apple and 1 chocolate bar.

How much do I need to pay?

A $3.00

B $3.50

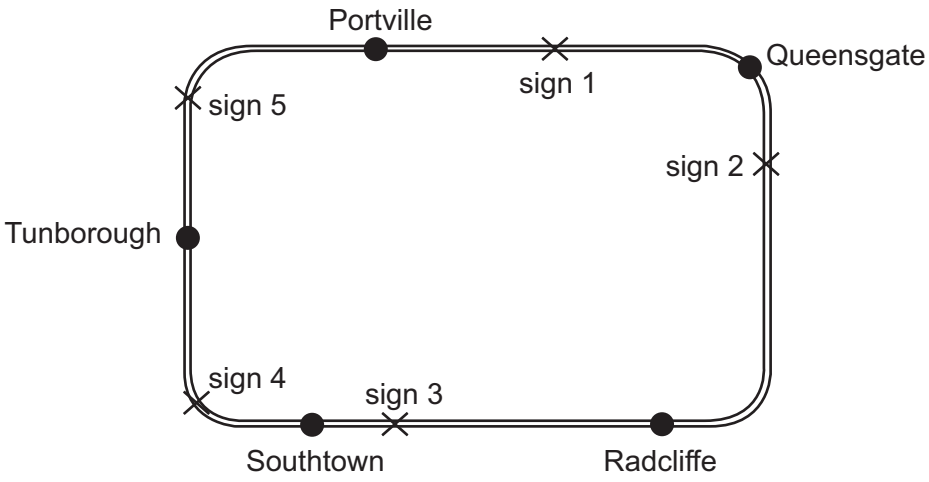
C $4.00

D $4.50

E $5.00

© Cambridge University Press & Assessment 2022

25 (Q50) There are five towns at various positions along a major ring road that encircles a city. Traffic travelling clockwise around the ring road is provided with information in the form of five 'distance' signs at various points along the road as shown.



The signs display the following information (all distances in km):

sign 1

Queensgate	10
Radcliffe	32
Southtown	49
Tunborough	64
Portville	88

sign 2

Radcliffe	17
Southtown	34
Tunborough	49
Portville	73
Queensgate	92

sign 3

Southtown	4
Tunborough	19
Portville	43
Queensgate	62
Radcliffe	84

sign 4

Tunborough	9
Portville	31
Queensgate	52
Radcliffe	74
Southtown	91

sign 5

Portville	13
Queensgate	32
Radcliffe	54
Southtown	71
Tunborough	86

One of the signs contains an error in one of the distances indicated.

Which one of the signs contains the error?

A sign 1

B sign 2

C sign 3

D sign 4

E sign 5

END OF TEST

© Cambridge University Press & Assessment 2022